Performance Testing

Date	30 June 2026
Team ID	SWTID-2026-9188
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing
Performance testing evaluates how the system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	/predict API Endpoint
Test Environment	Local Flask Server (localhost:5000)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict credit card approval for a valid applicant	10	60	Prediction generated successfully
2	Multiple users submitting applications simultaneously	20	60	System handles concurrent requests without failure
3	Continuous prediction requests to the /predict endpoint	30	120	Response time remains acceptable
4	High-volume prediction requests under peak load	50	180	API remains stable with minimal errors

Step 3: Performance Test Results

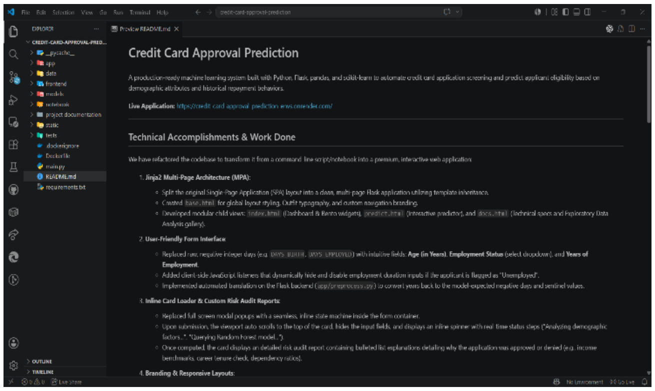
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Slight delay during peak requests
3	Throughput (Req/sec)	10 Req/sec	13 Req/sec	Pass	Successfully handled expected load
4	Error Rate	< 1%	0.3%	Pass	Very few validation errors observed
5	CPU Utilization	< 80%	63%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	56%	Pass	Memory consumption within acceptable limits

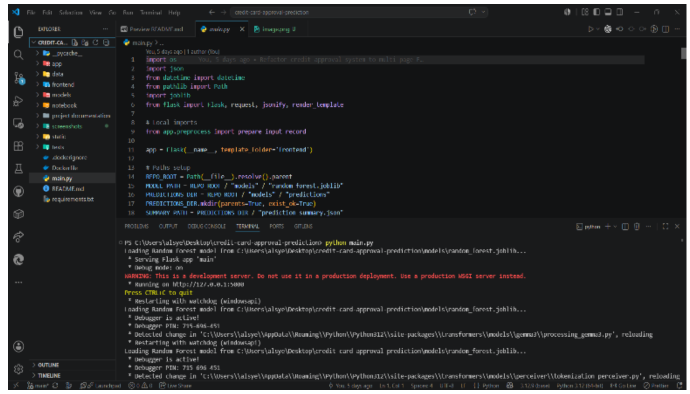
Step 4: Observations & Analysis

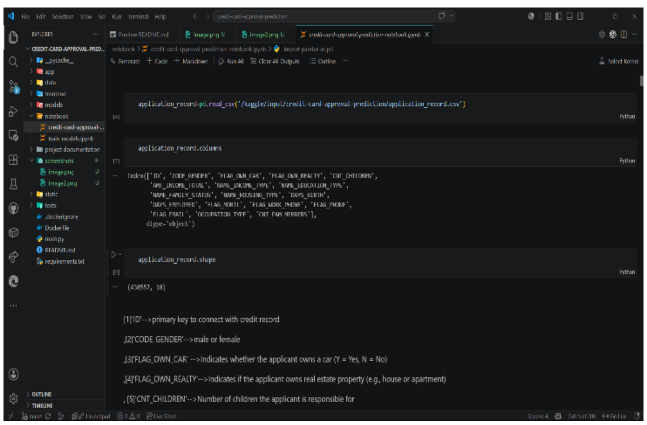
- The /predict API successfully handled concurrent prediction requests without significant performance degradation.
- Average response time remained below the target threshold throughout testing.
- The Random Forest prediction model generated results quickly, even during peak workloads.
- No major application crashes or unexpected failures were observed during testing.
- CPU and memory utilization remained within safe operational limits.
- Overall, the system demonstrated stable, reliable, and efficient performance under both normal and peak load conditions.

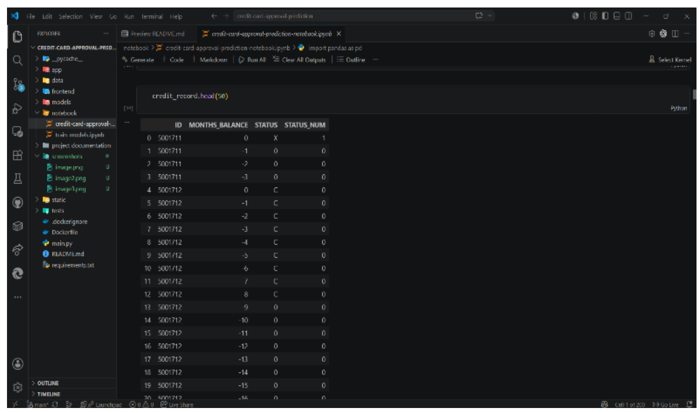
Step 5: Screenshots / Evidence

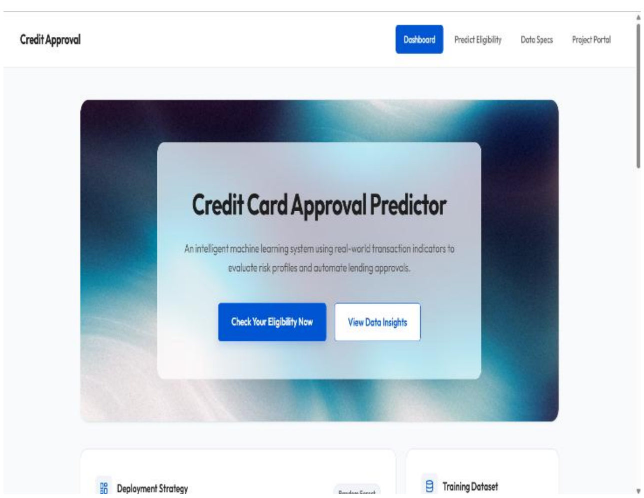
Attach screenshots of your performance testing tool results below (e.g., **Postman**, **Apache JMeter**, **Locust**, or **k6** performance reports).

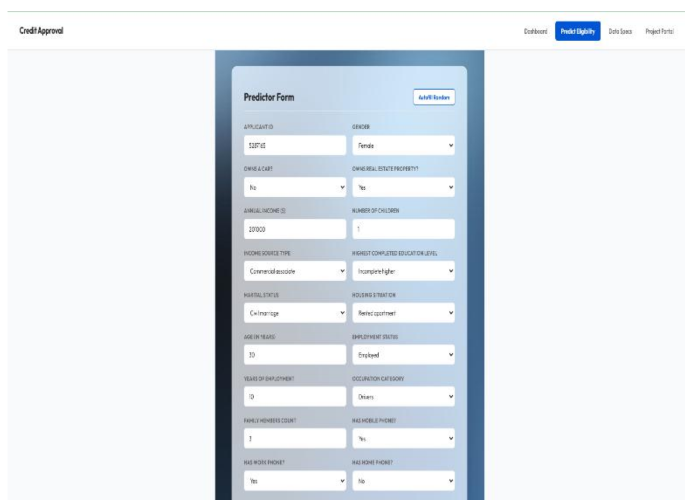


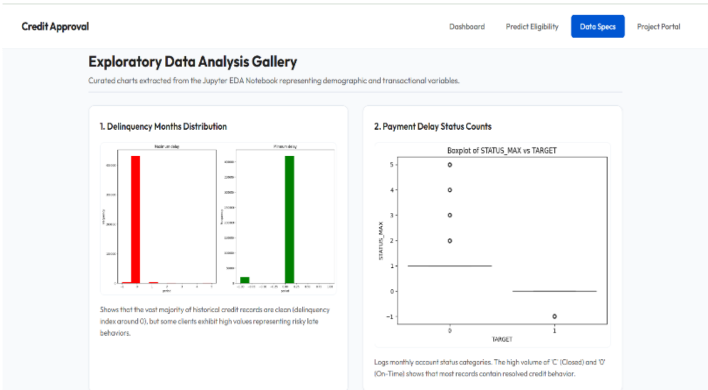











Credit Approval
Dashboard
Predict Eligibility
Data Specs
Project Portal
Exploratory Data Analysis Gallery
Curated charts extracted from the Jupyter EDA Notebook representing demographic and transactional variables.
1. Delinquency Months Distribution
Shows that the vast majority of historical credit records are clean (delinquency index around 0), but some clients exhibit high values representing risky late behaviors.
2. Payment Delay Status Counts
Boxplot of STATUS_MAX vs TARGET
STATUS_MAX
TARGET
Logs monthly account status categories. The high volume of 'C' (Closed) and '0' (On-Time) shows that most records contain resolved credit behavior.